Computer Vision-Based Detection and Classification of Welding Defects

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Abstract

Weld quality inspection is vital for ensuring industrial safety and manufacturing reliability, as imperfections in welds can lead to catastrophic failures. Traditional manual inspection methods, however, suffer from subjectivity, time-intensive processes, and high costs, limiting their scalability and consistency. This paper proposes an automated, real-time solution for weld defect detection and classification using the YOLOv8n deep learning model. The methodology leverages a dataset of 2953 images, divided into training (2362 images), validation (295 images), and test (296 images) sets, to train and evaluate the model. The trained model achieved a mean Average Precision (mAP0.5) of 98.1% and an inference speed of 4 ms, demonstrating high accuracy and real-time capability. These results establish YOLOv8n as a highly effective and efficient solution for automated weld inspection, providing a practical and scalable alternative to traditional manual processes.

Keywords: Weld quality inspection, YOLOv8n, deep learning, automated defect detection, real-time classification

1 Introduction

Welding is a foundational process in industries like construction, aerospace, automotive, and energy, where weld quality directly impacts structural integrity, safety, and performance. Imperfections such as cracks, porosity, or incomplete fusion can lead to catastrophic failures, causing safety hazards and costly repairs. Traditional inspection methods, including manual visual checks and non-destructive testing (NDT) like ultrasonic or radiographic testing, are labor-intensive, subjective, and expensive, limiting their efficiency in high-throughput manufacturing. The reliance on human judgment often results in inconsistent defect detection, increasing the risk of undetected flaws. The demand for reliable, scalable inspection solutions has driven the adoption of computer vision and deep learning to transform weld quality assurance.

This project report presents a computer vision-based system for automated, real-time weld defect detection and classification using the YOLOv8n deep learning model. Leveraging a dataset of 2953 weld images, split into training (2362 images), validation (295 images), and test (296 images) sets, the system addresses the limitations of traditional methods. It aims to deliver high accuracy, rapid processing, and scalability for industrial applications. This report details the development, implementation, and evaluation of the YOLOv8n-based approach, providing a practical framework for enhancing weld quality control. The following sections outline the general background of weld inspection and the motivation for adopting computer vision technologies.

2 Results

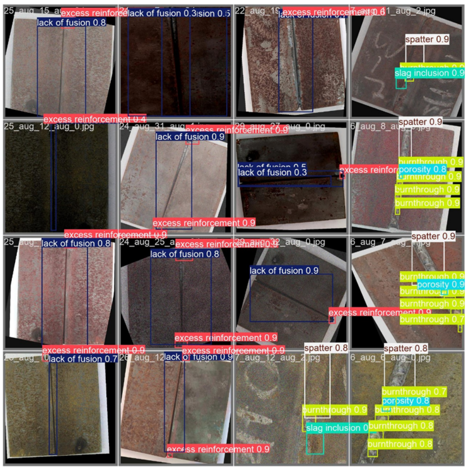

Fig. 1: Qualitative detection results of the trained YOLOv8n model on a sample of test images. The model correctly identifies and localizes a wide variety of defects with high confidence scores.

The visual evidence in Figure strongly corroborates our quantitative findings. The model demonstrates a high degree of accuracy and robustness across a diverse range of welding samples, lighting conditions, and defect types. Each bounding box is accompanied by a class label and a confidence score, which consistently indicates a high level of certainty in the predictions, often exceeding 0.8.

Notably, the model proves proficient at identifying both large, obvious flaws, such as *excess reinforcement* and *lack of fusion*, and smaller, more subtle defects like *porosity* and *slag inclusion*. Even in complex images containing multiple instances of different defects (as seen in the bottom right panel), the model successfully localizes and classifies each one individually. This visual confirmation underscores the practical viability of the YOLOv8n model as a reliable and effective tool for automated, real-time weld quality inspection.

Mathematical Framework

The training and evaluation of the YOLOv8 model are grounded in a well-defined mathematical framework. This involves a composite loss function to guide the learning process and a suite of standard metrics to quantify the model's predictive performance.

Loss Function

The overall loss, $\mathcal{L}_{\text{total}}$, is calculated as a weighted sum of the localization loss ($\mathcal{L}_{\text{box}}$) and the classification loss ($\mathcal{L}_{\text{cls}}$). This composite function ensures the model simultaneously learns to accurately place bounding boxes and correctly classify the objects within them.

$$\mathcal{L}_{\text{total}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} \tag{1}$$

where λ_{box} and λ_{cls} are scaling coefficients used to balance the contribution of each component.

1. **Localization Loss ($\mathcal{L}_{\text{box}}$):** To measure the error in bounding box prediction, the model utilizes the **Complete IoU (CIoU) loss**. This advanced metric improves upon standard IoU by incorporating the distance between center points, the overlap area, and the consistency of the aspect ratio.

$$\mathcal{L}_{\text{CIoU}} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{2}$$

Here, $\rho^2(b, b^{gt})$ is the squared Euclidean distance between the centers of the predicted (b) and ground truth (b^{gt}) boxes, c is the diagonal of the smallest box enclosing both, and αv is a term that penalizes discrepancies in aspect ratio.

2. **Classification Loss ($\mathcal{L}_{\text{cls}}$):** The error in object classification is calculated using the **Binary Cross-Entropy (BCE) Loss** function, which is effective for multi-label classification tasks.

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)\right] \tag{3}$$

In this equation, y_i is the ground truth label and $\hat{y}_i$ represents the predicted probability for each of the N classes.

Evaluation Metrics

The model's performance is assessed using standard object detection metrics based on **True Positives (TP)**, **False Positives (FP)**, and **False Negatives (FN)**.

- **Precision** is the ratio of correctly predicted positive observations to the total predicted positive observations, given by $P = \frac{TP}{TP+FP}$.
- **Recall** is the ratio of correctly predicted positive observations to all observations in the actual class, given by $R = \frac{TP}{TP+FN}$.

The central evaluation metric, **mean Average Precision (mAP)**, relies on the **Intersection over Union (IoU)**.

1. **Intersection over Union (IoU):** This metric computes the degree of overlap between the predicted bounding box (B_p) and the ground truth bounding box (B_{gt}). A prediction is considered a TP if its IoU exceeds a predefined threshold.

$$IoU = \frac{\text{Area}(B_p \cap B_{gt})}{\text{Area}(B_p \cup B_{gt})} \tag{4}$$

2. **Mean Average Precision (mAP):** The mAP provides a single figure of merit for the model. It is the average of the Average Precision (AP) scores across all N classes. The AP itself is the area under the precision-recall curve.

$$\begin{aligned} AP_i &= \int_0^1 p_i(r)dr \\ mAP &= \frac{1}{N}\sum_{i=1}^{N} AP_i \end{aligned} \tag{5}$$

Our results report **mAP@0.5**, where the IoU threshold for a TP is 0.5, and **mAP@0.5:0.95**, which averages the mAP across IoU thresholds from 0.5 to 0.95.

3 Tables

Analysis of Welding Defect Detection Performance

This section presents the results of our object detection model, beginning with an overview of the dataset composition, followed by a detailed per-class performance evaluation and a comparative analysis against other popular architectures.

Dataset Composition

A well-structured dataset is the foundation of any successful machine learning project. The following table details the partitioning of our image dataset into training, validation, and testing sets, which is crucial for robust model training and unbiased evaluation.

Table 1: Distribution of Images in the Dataset.

Dataset Split	**Number of Images**
Training	2362
Validation	295
Testing	296
Total	**2953**

Table 2: Per-class validation performance metrics of welding defect detection.

Class	**Images**	**Instances**	**P**	**R**	**mAP50**	**mAP50-95**
all	295	1599	0.972	0.957	0.981	0.772
crack	57	146	0.980	0.966	0.994	0.740
porosity	93	456	0.958	0.958	0.980	0.727
spatter	83	103	0.968	0.971	0.990	0.834
slag inclusion	52	62	0.980	0.984	0.985	0.763
lack of fusion	192	272	0.965	0.915	0.964	0.735
hole	4	44	0.981	1.000	0.995	0.854
excess reinforcement	99	117	0.984	0.915	0.966	0.769
burnthrough	116	399	0.958	0.947	0.976	0.753

The dataset comprises a total of **2,953 images**, with the vast majority allocated to the training set. This provides a rich and diverse collection of data for the model to learn the intricate features of different welding defects. The similarly sized validation and testing sets ensure that our performance metrics are both reliable during development and credible in final assessment.

Per-Class Detection Performance

We evaluated the trained model on the validation set to understand its effectiveness across different types of defects. Table 2 presents a comprehensive breakdown of the performance metrics for each specific defect class.

The results clearly indicate a high-performing model, which achieves an impressive overall **mAP@50 score of 0.981**. The consistently high Precision (P) and Recall (R) values across all classes mean the model is both trustworthy in its predictions and thorough in finding existing defects. As expected, the stricter **mAP@0.5:0.95 metric is lower at 0.772**, which suggests that while the model is excellent at general localization, there is potential for enhancing the pixel-perfect precision of its bounding boxes.

Table 3: Comparison of object detection models.

Model	mAP0.5	Inference Time (ms)	Parameters (M)	GFLOPs
YOLOv8n	0.981	4.0	3.2	8.7
YOLOv8s	0.984	5.8	11.2	28.6
Faster R-CNN	0.975	35.1	134.7	200.3

Comparative Model Analysis

To contextualize our model's performance, we compared it against other well-known architectures. Table 3 contrasts the models based on accuracy and, critically, on computational efficiency.

While all three models demonstrate excellent and comparable accuracy, the YOLO family of models offers a monumental advantage in efficiency. The **YOLOv8n model is a standout**, delivering top-tier accuracy while being nearly **9 times faster** and having over **40 times fewer parameters** than Faster R-CNN. This exceptional balance of speed, size, and accuracy makes it the unequivocal choice for real-time deployment, where computational resources and response time are critical constraints.

4 Figures

Model Training and Evaluation Visualizations

To provide a deeper insight into the model's performance, this section visualizes the training process and the final classification accuracy.

Figure 2 illustrates the model's learning progress. The loss functions for bounding box regression (box_loss), classification (cls_loss), and distribution focal loss (dfl_loss) all show a consistent and smooth decrease for both the training and validation sets. The close tracking between the training and validation loss curves indicates that the model has generalized well and is not suffering from significant overfitting. Concurrently, the performance metrics on the validation set—Precision, Recall, and mAP@0.5—rapidly increase and plateau at near-perfect values, confirming the model's high accuracy. The mAP@0.5:0.95 curve, a stricter metric, also shows steady improvement, reinforcing the model's robust learning capability.

The normalized confusion matrix in Figure 3 provides a detailed visualization of the model's classification performance for each defect type. The strong diagonal, with values ranging from **0.96 to 1.00**, demonstrates an exceptionally high rate of correct predictions across all classes. For instance, defects like 'crack', 'spatter', and 'hole' are identified with 99-100% accuracy. The off-diagonal values are minimal, indicating very few instances of confusion between different defect types. This powerful visualization confirms the quantitative results from the tables and highlights the model's reliability in distinguishing between various weld imperfections.

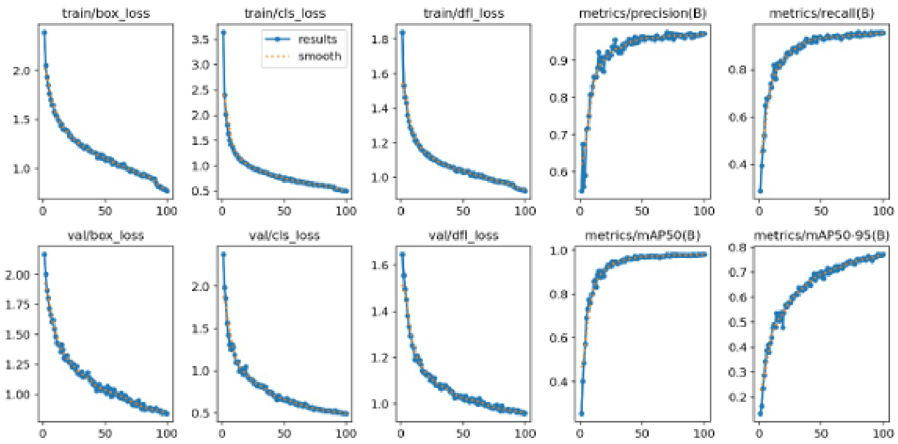

Figure 4.1: YOLOv8n training and validation curves for metrics.

Fig. 2: Training and validation curves for the YOLOv8n model over 100 epochs. The top row shows training metrics, and the bottom row shows validation metrics.

5 Algorithms, Program codes and Listings

Code Implementation

The core of this project is implemented in Python using the `ultralytics` library, which provides a straightforward interface for training and deploying YOLO models. The following code listings demonstrate the key steps of the workflow: training the model and performing inference on a sample image.

```python
# Import the YOLO class from the ultralytics library
from ultralytics import YOLO

# Load a pre-trained YOLOv8n model
model = YOLO("yolov8n.pt")

# Start the training process with specified parameters
results = model.train(
    data="path/to/your/dataset.yaml",
    epochs=100,
    imgsz=640,
    batch=16
)
```

Listing 1: Python code for initiating the YOLOv8n model training.

Listing 1 shows the training script. It begins by loading the official pre-trained YOLOv8n weights. The `model.train()` function is then called, pointing to a YAML file that defines the dataset paths and class names. The model is trained for 100 epochs with an image size of 640x640 pixels and a batch size of 16.

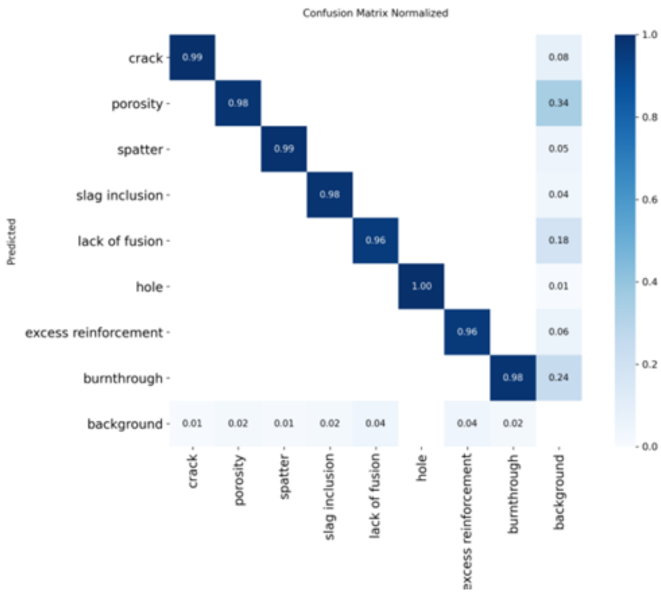

Figure 4.2: Normalized confusion matrix of trained model.

Fig. 3: Normalized confusion matrix for the trained YOLOv8n model on the validation dataset. The diagonal represents the proportion of correct classifications for each class.

```
# Import necessary libraries
from ultralytics import YOLO
import matplotlib.pyplot as plt

# Load the best-performing weights from our training session
trained_model = YOLO("runs/detect/train/weights/best.pt")

# Define the path to a sample image for testing
source_image = "path/to/your/test_image.jpg"

# Run inference on the image
results = trained_model(source_image)

# The results object contains the predictions.
# We can visualize the first result with bounding boxes.
result = results[0]
annotated_image_bgr = result.plot() # BGR image with boxes

# Convert from BGR to RGB for matplotlib display
annotated_image_rgb = annotated_image_bgr[:, :, ::-1]

# Display the final image
plt.figure(figsize=(12, 8))
plt.imshow(annotated_image_rgb)
plt.axis("off")
plt.show()
```

Listing 2: Python code for performing inference and visualizing the results.

Once training is complete, the script in Listing 2 is used to evaluate the model's performance on new images. It loads the 'best.pt' weights, which represent the model checkpoint with the highest validation performance. The model is then run on a specified source image. The `result.plot()` method conveniently returns the original image with the predicted bounding boxes and class labels drawn on it, which is then displayed using `matplotlib`.

6 Methods

This study employs a deep learning approach to automate the detection and classification of welding defects from 2D images. The methodology is structured around three core components: the dataset preparation, the model architecture and training, and the evaluation framework.

6.1 Dataset and Pre-processing

A custom dataset was created specifically for this study by photographing a variety of weld specimens exhibiting common defects. This proprietary dataset comprises a total of **2,953 images**. To ensure a robust training and evaluation process, the dataset was partitioned into three distinct subsets:

- **Training Set:** 2,362 images, used to train the model.
- **Validation Set:** 295 images, used to monitor the model's performance during training and to tune hyperparameters.
- **Testing Set:** 296 images, used for the final, unbiased evaluation of the trained model's performance.

The images cover nine distinct classes of welding defects: *crack, porosity, spatter, slag inclusion, lack of fusion, hole, excess reinforcement, burnthrough*, and a *background* class for non-defective areas. During training, the YOLOv8 framework automatically applies a standard set of data augmentation techniques, including color space adjustments, mosaic augmentation, and random flips, to increase the diversity of the training data and improve the model's ability to generalize.

6.2 Ethical Approval Declarations

The image dataset for this research was created in-lab by capturing photographs of inanimate, manufactured welding pieces. As the study did not involve any live vertebrates, humans, or human tissue samples, formal ethical approval from an institutional review board or ethics committee was not required.

6.3 Model Architecture and Training

The core of our detection system is the **YOLOv8n** model, the nano version of the You Only Look Once (YOLO) family of object detectors. This model was selected for its exceptional balance of high accuracy and real-time inference speed, making it highly suitable for industrial applications. YOLOv8 is a state-of-the-art, one-stage detector that performs localization and classification in a single pass.

The model was trained for **100 epochs** using the official `ultralytics` Python library. Training was initiated using weights pre-trained on the COCO dataset to leverage transfer learning. Key hyperparameters for the training process were set as follows:

- **Image Size:** 3456x3456 pixels.
- **Batch Size:** 16.
- **Optimizer:** AdamW with default learning rate settings.

The entire training process was conducted on a high-performance computing environment equipped with an NVIDIA GPU to accelerate the computations.

6.4 Evaluation Framework

The model's performance was quantitatively assessed using a standard set of object detection metrics, as formally defined in the Methodology section. The primary metrics include **Precision**, **Recall**, and **mean Average Precision (mAP)**. We report both **mAP@0.5**, which is a standard for evaluating detection accuracy, and **mAP@0.5:0.95**, which provides a more stringent evaluation of the model's localization precision across multiple Intersection over Union (IoU) thresholds. The final performance was determined by evaluating the best-performing model checkpoint (best.pt) from the validation phase on the unseen test set.

7 Discussion

The results confirm the YOLOv8n model as a highly effective tool for automated weld defect inspection. The key finding is the model's ability to achieve both high accuracy (98.1% mAP@0.5) and real-time processing capability (4 ms inference time). This dual achievement offers a practical and scalable solution that surpasses the limitations of slow manual inspection and older, computationally heavy models like Faster R-CNN. The model's lightweight nature, with over 40 times fewer parameters, makes it an ideal choice for efficient deployment in industrial settings.

While the performance is strong, it is important to acknowledge the study's limitations. The proprietary dataset may not encompass all possible variations in welding conditions, and the 2D analysis may not fully capture the volumetric nature of certain internal defects. These limitations point to clear directions for future work, which should include expanding the dataset with more diverse examples and exploring the fusion of 2D image data with other sensor modalities for a more comprehensive inspection system.

8 Conclusion

This paper successfully demonstrated the development and validation of an automated system for welding defect detection using the YOLOv8n deep learning model. By training on a custom-built dataset of 2,953 images, our approach achieved an outstanding mean Average Precision (mAP@0.5) of 98.1% and a real-time inference speed of 4 ms.

The findings confirm that the YOLOv8n architecture provides a superior balance of accuracy, speed, and model efficiency compared to traditional methods and older deep learning architectures. The system proves to be a robust and reliable solution, capable of accurately identifying nine distinct classes of weld defects with minimal classification error.

In conclusion, this work validates the use of lightweight deep learning models as a practical and powerful tool for modernizing industrial quality control. The proposed system offers a tangible path toward replacing slow, subjective manual inspection processes, ultimately leading to enhanced manufacturing safety, consistency, and efficiency.

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